ROOT4: an original-law physical bridge

Research audit note

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1 The original-law bridge for ROOT4

This note proves a conditional analytic implication, not the terminal ROOT4 inequalities or their implementation. The sole smoothing input is the closed-half-line Prawitz majorant stated below. All other steps in this note are proved for the original iid marginal, including laws with infinite support inside a bounded interval and atoms at zero or at a threshold. No optimizer or contact assumption is used.

Let X have mean zero, variance one, $|X| \le R = 17/4$, and $\rho = \mathbb{E}|X|^3 < 46/25$. Set

$$u = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad v_\pm = \mathbb{E}X_\pm^2, \quad a = v_-/u, \quad b = v_+/u, \quad w = u/a + u/b, \quad \vartheta = v_+ - v_-.$$

Both $v_\pm$ and a, b are positive; also $a, b \le R$. Signwise Cauchy gives $u/a \le \mathbb{P}(X < 0)$ and $u/b \le \mathbb{P}(X > 0)$, so $0 < w \le 1$. The canonical probability

$$Q = (u/a)\delta_{-a} + (1-w)\delta_0 + (u/b)\delta_b$$

has the same first and second moments on each strict half-line as P. Put $r = \rho(Q)$ and $\eta = \rho - r$.

1.1 One actual measure, including all contact atoms

For $y = |X|$, let $c = b$ on the positive tag and $c = a$ on the negative tag, and put $W = y(y-c)^2$ (zero at $X = 0$). Direct expansion gives

$$\mathbb{E}W = \eta \ge 0, \tag{1}$$

$$\mathbb{E}X^3 = b - a + \mathbb{E}[\operatorname{sgn}(X)W],$$

$$\mathbb{E}X^4 = a^2 - ab + b^2 + \mathbb{E}[(y+2c)W]. \tag{2}$$

Indeed $y^3 = W + 2cy^2 - c^2 y$ and $y^4 = (y+2c)W + 3c^2y^2 - 2c^3 y$; integrate on each tag using $\mathbb{E}[y; \text{tag}] = u$ and $\mathbb{E}[y^2; \text{tag}] = uc$. If $\eta = 0$, then P is supported on $\{-a, 0, b\}$ and its first two moments force $P = Q$. This case requires no defect probability.

For $\eta > 0$, define the single tagged probability $\nu = W\,dP/\eta$. It charges neither zero nor either contact. On its domain set

$$k_0 = (c^2 y)^{-1}, \qquad k_2 = (y-c)^{-2}, \qquad k_1 = (y-c)^{-1}.$$

Then, writing $p_0 = 1 - w$,

$$\eta \int k_0\, d\nu = p_0 - P\{0\},$$

$$\eta \int_\sigma k_2\, d\nu = u - c_\sigma P\{\sigma c_\sigma\}, \qquad \int_\sigma k_1\, d\nu = 0. \tag{3}$$

To check the first formula, $Wk_0 = (y-c)^2/c^2$ off zero and contacts; its integral on a tag is $P(\text{tag}) - u/c$. For k_2, cancellation gives y only away from the contact, so the omitted contact

mass must be subtracted. For k_1 it gives $y(y-c)$, whose integral is zero even after omitting the contact. All integrals exist; for k_1, Cauchy uses the finite k_2 integral.

Let $\psi(q) = \mathbb{E}e^{\mathrm{i}qX}$ and $g(q) = \int e^{\mathrm{i}qx}\,dQ(x)$. Hermite interpolation of $f(y) = e^{\mathrm{i}\sigma qy}$ at $0, c, c$ gives

$$f(y) - L_f(y) = y(y-c)^2 r_q(\sigma, y), \qquad r_q = f[0, c, c, y].$$

The interpolant is quadratic with constant $f(0) = 1$. Its linear and quadratic integrals match on each tag; its constants match only after both tags and the zero atoms are combined. Consequently

$$\psi(q) - g(q) = \eta \int r_q\,d\nu. \tag{4}$$

This identity uses the same measure at all frequencies. The integral form of a third divided difference gives $|r_q| \le |q|^3/6$. For completeness another bound is $|r_q| \le q^2/y$: write

$$L_f(y) = f(c) + f'(c)(y-c) + \frac{f(0) - f(c) + cf'(c)}{c^2}(y-c)^2.$$

The quadratic coefficient and the first-order Taylor remainder at c are bounded by $q^2/2$ and $q^2(y-c)^2/2$, respectively. Division gives the claim. Thus the following three radii are valid:

$$|\psi(q) - g(q)| \le \min\{\eta|q|^3/6,\ \max(a,b)^2 p_0 q^2,\ 1 + |g(q)|\}. \tag{5}$$

The mass radius follows from $1/y = c^2 k_0$ and (3).

For arbitrary finite coefficients κ_j, define $A = \Re\sum_j 4\kappa_j g(q_j)^3 r_{q_j}$. If fixed prices $\lambda_0, \lambda_+, \lambda_- \ge 0$ and signed $\theta_\pm$ satisfy

$$A - \lambda_0 k_0 - \lambda_\sigma k_2 - \theta_\sigma k_1 \le S$$

on the entire tagged domain, then

$$\Re\sum_j 4\kappa_j g(q_j)^3(\psi(q_j) - g(q_j)) \le p_0\lambda_0 + u(\lambda_+ + \lambda_-) + S\eta =: \Gamma(\eta). \tag{6}$$

This holds for each actual P, not for a purported sufficiency characterization of all measures satisfying the budgets. Enlarging that class in an upper-bound problem is safe. Price choices must precede the supremum; signed equality prices and a negative S are retained. At $\eta = 0$ the response is zero and the displayed right side is nonnegative, so the same bound applies directly.

1.2 The physical smoothing and its two reserves

Let $Z = (X_1 + \cdots + X_4)/2$ with independent copies of this same law. The input Prawitz inequality is, for every real x,

$$\mathbb{P}(Z \le x) \le J_P(x) := \frac{1}{2} + \int_0^1 \Re\{\widehat{K}(t)e^{-6\mathrm{i}tx}\psi(3t)^4\}\,dt, \quad \widehat{K}(t) = 1 - t + \mathrm{i}[(1-t)\cot(\pi t) + 1/\pi]. \tag{7}$$

Only the real integrand is integrated. Put $S(t) = t\Im\widehat{K}(t)$. Its endpoint-continuous extension satisfies

$$0 \le S \le 1/3, \qquad |S'| < 2, \qquad |S''| < 5. \tag{8}$$

Here are explicit sufficient bounds. The cotangent partial fractions give

$$S = \pi^{-1} - 2\pi^{-1}(A + MG), \quad A = t^2/(1+t),\ M = t^2(1-t),\ G = \sum_{j\ge 2}(j^2 - t^2)^{-1}.$$

On $[0,1]$, $|A'| \le 3/4$, $|A''| \le 2$, $|M| \le 1/6$, $|M'| \le 1$, $|M''| \le 4$, and $G \le 3/4$, $|G'| \le 10/27$, $|G''| \le 106/135$. The first sum telescopes; the latter two follow from $j^2 - t^2 \ge 3j^2/4$ and

$\sum_{j\ge2} j^{-k} \le 2^{-k} + \int_2^\infty s^{-k}\,ds$. The derivative series converge uniformly. With $\pi > 3$, these imply $|S'| < 19/18$, $|S''| < 223/54$. Nonnegativity follows directly from $(1-t)\cot(\pi(1-t)) \le 1/\pi$ when $t > 1/2$, and from the nonnegative cotangent when $t \le 1/2$. The positive partial-fraction terms give $S \le 1/\pi < 1/3$.

Set $D = x - Z$, $A_x(t) = \mathbb{E}\cos(6Dt)$ and $L_x(t) = \mathbb{E}[6D\,\mathrm{sinc}(6Dt)]$. The regular real integrand is $(1-t)A_x(t) + S(t)L_x(t)$. Since $\mathrm{sinc}(v) = \int_0^1 \cos(sv)\,ds$,

$$|A_x'| \le 6m_1,\ |A_x''| \le 36m_2, \quad |L_x| \le 6m_1,\ |L_x'| \le 18m_2,\ |L_x''| \le 72m_3, \quad m_j = \mathbb{E}|D|^j.$$

Differentiation under these bounded-moment integrals and the product rule therefore give

$$|F_x''(t)| \le 42m_1 + 108m_2 + 24m_3. \tag{9}$$

All four copies are used to obtain

$$\mathbb{E}D^2 = x^2 + 1 =: D_2, \qquad \mathbb{E}D^4 = x^4 + 6x^2 - 2x\mathbb{E}X^3 + (\mathbb{E}X^4 + 9)/4.$$

Thus for valid whole-cell moment bounds $\kappa_* \ge |\mathbb{E}X^3|$ and $M_* \ge \mathbb{E}X^4$, put $D_4 = x^4 + 6x^2 + 2|x|\kappa_* + (M_* + 9)/4$. Then $m_1 \le \sqrt{D_2}$ and $m_3 \le \sqrt{D_2 D_4}$. The bounds (2) supply these without an additional fourth-moment premise. On $|x| \le 2$, $m_2 \le 5$, $\mathbb{E}D^4 < 54$, $m_1 < 9/4$, $m_3 < 33/2$, so (9) is less than $2061/2$.

Choose $N \ge 1$, $t_j = (j+1/2)/N$, $q_j = 3t_j$ and $\kappa_j(H) = \widehat{K}(t_j)e^{-\mathrm{i}q_j H}/N$. Composite midpoint quadrature has error at most $\sup|F''|/(24N^2)$. Consequently the raw numerator $2(J_P(H/2) - \Phi(H/2))$ has the single reserve

$$Q_H = \min\left\{\frac{687}{8N^2}, \frac{42\sqrt{D_2} + 108D_2 + 24\sqrt{D_2 D_4}}{12N^2}\right\}. \tag{10}$$

This quadrature is for the original law once, before reference expansion. Neither the reference nor its response/remainder needs another quadrature.

To control threshold joins, suppose $a_j \ge |g(q_j)|$ and e_j bounds $|\psi(q_j) - g(q_j)|$ uniformly on the entire excess cell. The actual characteristic function is 1-Lipschitz, since $\mathbb{E}|X| \le 1$. Hence

$$P_j = \min(1, a_j + e_j + 3/(2N))$$

bounds $|\psi(3t)|$ throughout its frequency cell. Define d_j as the smaller of the two integrals over that cell of

$$t^2(1-t) + t/3, \qquad t^2(1-t) + t(1-t)/2.$$

Both dominate $t^2|\widehat{K}(t)|$: use $|\widehat{K}| \le 1 - t + \Im\widehat{K}$ and, for the second, $S \le (1-t)(1+2t)/(\pi(1+t)) \le (1-t)/2$. Their full integrals are $1/4$ and $1/6$. Thus

$$|J_P''(x)| \le B := 36\sum_j d_j P_j^4.$$

For the smooth physical function $G_P(H) = 2(J_P(H/2) - \Phi(H/2))$, $|G_P''| \le (B + 1/4)/2$, since $\sup|\Phi''| < 1/4$. On any raw-threshold interval of length h, its chord lies above G_P up to the single reserve

$$E_H = h^2(B + 1/4)/16. \tag{11}$$

There is no differentiation or interpolation of the CDF. The moment reserve (10) may be computed separately at each node; the curvature reserve is common to the whole geometry/excess cell.

1.3 A precise conditional certificate implication

Theorem 1.1 (Low-excess original-law bridge)**.** *Fix a geometry cell, an excess interval $[L,U]\subset[0,\infty)$, and raw nodes $H_0<\cdots<H_m$ of spacing at most h. Assume all the following bounds are valid for every actual standardized P in this cell, with $\rho=r+\eta<46/25$ and $\eta\in[L,U]$. At each node let $M_H\ge C_H(Q)-Tr$, where*

$$C_H(Q)=1+2\Re\sum_j\kappa_j(H)g(q_j)^4-2\Phi(H/2).$$

Let $\Gamma_H(\eta)$ be a whole-domain price bound (6)*, and let $R_{2,H}(\eta)$ bound*

$$\sum_j|\kappa_j(H)||\psi(q_j)^4-g(q_j)^4-4g(q_j)^3(\psi(q_j)-g(q_j))|.$$

Suppose the complete row

$$P_H(\eta)=M_H+Q_H+E_H+2\Gamma_H(\eta)+2R_{2,H}(\eta)-T\eta$$

is a convex quadratic, nonpositive at L and U, at every node. Then $2\{\mathbb{P}(X_1+\cdots+X_4\le H)-\Phi(H/2)\}\le T\rho$ for every $H\in[H_0,H_m]$ and every such law, including $\eta=0$.

Proof. Expand the actual fourth power at each frequency. The linear part is controlled by the one common measure, and the remaining complex part by its full nonnegative modulus bound. Midpoint quadrature yields $G_P(H)-T\rho\le P_H(\eta)-E_H$ at each node. Convexity gives $P_H(\eta)\le 0$ on $[L,U]$. The chord inequality (11) now implies $G_P(H)-T\rho\le 0$ between nodes. Finally use (7). All these inequalities hold for one arbitrary law throughout; no law is reconstructed from a relaxation. □

An elementary permitted remainder is obtained from

$$\psi^4-g^4-4g^3(\psi-g)=(\psi-g)^2(\psi^2+2\psi g+3g^2).$$

If $|g|\le a\le 1$ and $|\psi-g|\le A+B\eta$ nonnegatively on the whole cell, freeze $p=\min(1,a+\max(A+BL,A+BU))$. Then $(A+B\eta)^2(p^2+2pa+3a^2)$ is a convex quadratic remainder. Other complete bounds, including a proven disk/lens bound, may replace it. Their correctness remains a premise if not proved separately. Selecting a whole valid row independently at each node is safe; interpolating a pointwise minimum of candidate curves is not needed.

For $H\le -4$, Cantelli gives $F_Z(H/2)\le 1/5$. For $H\ge 25/8$, the elementary normal bound $\Phi(25/16)>4/5$ gives the same raw numerator bound $2/5$. Since $\rho\ge 1$, both positive-score tails are at most $2/5<T=4097/10000$. To verify that normal bound directly, $\pi<22/7$ gives $\phi(0)>997/2500$ by squaring. With $x=25/16$, the exponential Taylor lower bound gives

$$\Phi(x)\ge\frac{1}{2}+\frac{997}{2500}\left(x-\frac{x^3}{6}+\frac{x^5}{40}-\frac{x^7}{336}\right)>\frac{4}{5}.$$

1.4 The independent count route and reflection

Let $p=P(X\ge 0)$, $q=1-p$. Both are positive. Conditioning on the number j of nonnegative observations gives mass $\binom{4}{j}p^jq^{4-j}$. Conditional on any fixed sign pattern, independence retains independent conditional marginals. Thus Z has conditional mean and variance

$$\mu_j=\frac{u(j-4p)}{2pq},\qquad V_j=\frac{1}{4}\left[j\left(\frac{v_+}{p}-\frac{u^2}{p^2}\right)+(4-j)\left(\frac{v_-}{q}-\frac{u^2}{q^2}\right)\right].$$

The same moments hold after mixing patterns with that count. Closed-event Cantelli yields the source bound $A_j(x)=1$ for $x\ge\mu_j$ and $A_j(x)=V_j/(V_j+(\mu_j-x)^2)$ otherwise. This includes

$V_j = 0$. Cauchy gives $w(1-\vartheta)/2 \le p \le 1 - w(1+\vartheta)/2$. Taking upper bounds on all five A_j on a cell and then a valid upper bound of their single binomial mixture preserves the original-law bound. No arbitrary dependent sign allocation can replace the iid weights.

Any complete count-based above-cut bound, or independent all-law large-ρ bound, can join the low-excess theorem when both closed ranges cover their common cutoff. A parent geometry may be replaced by all four closed children, each with its own complete low/high cover. No unexecuted terminal inequality is supplied by this observation.

For the reflected law $\widetilde{P} = \mathcal{L}(-X)$, a, b are swapped, ϑ changes sign, η and ρ are unchanged, and

$$\widetilde{p} = P(X \le 0) = 1 - p + P\{0\}, \qquad \widetilde{\psi}(q) = \overline{\psi(q)}.$$

The raw phase $e^{-\mathrm{i}qH}$ is retained when evaluating its own threshold H. Applying both positive closed bounds covers the negative CDF discrepancy because

$$\Phi(x) - F_Z(x) \le \Phi(x) - F_Z(x-) = F_{-Z}(-x) - \Phi(-x).$$

Equality at an atom is not assumed. These facts supply the analytic coverage implication only. Complete numerical enclosures, their selected implementation, and the unrestricted-owner localization remain separate.